

ings with the ministers. There are many challenges that India needs to resolve to build and develop a fab ecosystem.

First, a 'fab'—a term used to describe a semiconductor manufacturing facility—needs to be in a place with lots of space. For reasons of economics, it is more cost-efficient to build a cluster of fabs close to each other. This location should have a very good supply of water since fabs are big water-guzzlers. Also, the electric supply must be available 24x7, uninterrupted, 365 days a year—except for scheduled maintenance. The power should be free of any glitches, since some of the semiconductor equipment used for building today's chips require glitch-free power. The location should be in a place protected from natural disasters, especially earthquakes that can, unfortunately, irreparably damage billions of dollars of fab and equipment. In India, where the infrastructure is still not fully developed, and customs clearance can sometimes take weeks, it is imperative that the airport and customs for key components needed for export outside of India be placed right next to the cluster of fabs and the assembly, testing and packaging facilities.

Moreover, the zones where semiconductor fabs and back-end [assembly, testing and packaging] are located should be by law protected from strikes and unnecessary time-off provisions to enable competitiveness in the global markets.

Also, since a state-of-the-art fab can take up to five years before going into full production, not only full financing has to be in place to complete the job on time, but the labour should also be deployed through 365 days round the clock to make sure the fab is ready as quickly as possible. Fabs are very capital-intensive by nature and generating profits fast is an essential consideration in their implementation.

Beyond locating and building structures, we will need a variety of high-purity gases and wafers to fabricate the chips. Initially, we will have to import many of these raw materials from sources outside of our country, but with time, we can bring some of these supplies in-house.

An even bigger challenge than building a fab is 'filling a fab'—we need a very high volume of chips to fully utilise the fab capacity.

ON CHALLENGES OF A FAB ECOSYSTEM

A bigger challenge than building a fab is 'filling a fab'—we need a very high volume of chips to fully utilise the fab capacity. If a fab is not run at over 90 per cent-plus capacity round the clock, the chip costs will go up because of poor capacity utilisation, rendering our products uncompetitive.

ON PICKING THE RIGHT TECH

The choice of picking the right tech node is both a function of the products we choose to initially fill our domestic fab with, and where the performance, power and cost needs of these functions will be in the coming years and decades.

ON BUILDING A FAB ECOSYSTEM

We need to encourage local businesses to build domestic companies like Xiaomi, Oppo, Nokia, Huawei, Lenovo, and Dell. These OEMs should be supported by a supply of locally designed chips. Thus, we also need to create fabless design companies in India like MediaTek, Broadcom, Qualcomm and Marvell.

If a fab is not run at over 90 per cent-plus capacity round the clock, the chip costs will go up because of poor capacity utilisation, rendering our products uncompetitive.

Q: In your opinion, what technology or nodes should India look at to begin its fab journey?

A: The technology node (also called process node) refers to a specific set of process design rules that enable the technology to shrink to the next viable generation. The smaller the node, the smaller the feature size, producing smaller transistors that are both faster and more power efficient. But the cost of building chips is higher as the cost of equipment for defining the smaller dimensions skyrockets. Over the past 50 years, the process technology has evolved over a score times. Thanks to Moore's Law, an observation made in the 1960s by my guru Gordon Moore—the co-founder of Intel—we have been able to shrink the size of chips by half every two to three years.

The choice of picking the right technology node is both a function of the products we choose to initially fill our domestic fab with, and where the performance, power and cost needs of these functions will be in the coming years and decades. Also, keep in mind that our domestic fab capacities will not only be filled with products we need for our domestic needs but hopefully also supply some of the export needs of the rapidly growing global semiconductor market that is projected to double from around \$500 billion now [that it took 50 years to reach] to \$1 trillion in the next decade.

Today the fabrication of chips has progressed from nodes between 180 nanometres (nm) with the older technology to 3-5 nm with the most advanced technologies used to build high-performance computers for cloud computing. For the bulk of India's needs in the future, 28 nm appears to be the sweet spot, but for now, anything starting with 65 nm and below will be a good start. For example, the strongest-growing segment in the coming decade is likely to be automotive, where demand is projected to triple over the next decade. It is estimated that the bulk of the automotive needs can be met with technologies at 65 nm and below.

However, there is a lot of heavy lifting